

WALKTHROUGH • AGENTIC TRIAL OPERATIONS ON THE HYBRID STACK

The Hybrid AI Trial-Operations Layer

75 governed automations across project management, trial management, and monitoring — on Claude + local.

AI drafts • triages • routes. Validated tools + humans own every regulated number and record.

Trial operations are fragmented

Spread across systems

Knowledge and work live in the EDC, eTMF, Smartsheet, Pinnacle 21, email, and a dozen SOPs — none of it joined up.

Manual busywork

QC triage, document filing, status decks, action-item chasing — repetitive, cross-system, and endless.

High stakes

It's regulated work — so any automation has to be governed, not a free-for-all chatbot.

Exactly the work an AI layer can absorb — if, and only if, it's governed properly.

Point the hybrid stack at operations

75 governed agents

Across PM, trial management, document control, data QC, and risk-based monitoring.

Routed by sensitivity

PHI / unblinded → local model; de-identified reasoning → Claude. The gateway decides.

Tiered by risk

Green drafts • amber triage with verify • red stays with validated tools + humans.

Gated by a human

Every regulated change is proposed to an approval queue, with a Part 11 audit trail.

One gateway, many agents

Orchestration scheduler + agent framework — fires agents on schedule or study events



The Gateway (LiteLLM-class) data-classification router · Presidio PHI egress firewall · cost/audit · caching · one Part 11 trail



Local model sensitive / PHI / unblinded

Claude API (BAA) de-identified reasoning

RAG knowledge base SOPs / WIs / study docs



connectors (MCP)



Systems of record eTMF · Smartsheet · EDC · CTMS · SharePoint · Outlook · Pinnacle 21 (read-mostly; writes gated)



Human-in-the-loop approval queue every regulated change proposed here — AI rationale + sources — for a human to approve / edit / reject

Why this is safe to run

GREEN · 14

Drafts, summaries, cited RAG answers — advisory only.

AMBER · 43

Classification, triage, code, proposed changes — with a human verify before commit.

RED · 18

Reported numbers, records of truth, unblinded data — validated tool + human; LLM out of the path.

The three hard rules

- 1. PHI / unblinded / randomization data never leaves**
always the local frozen model (zero egress); de-identified text may use Claude.
- 2. Every reported number & record of truth comes from a validated tool**
Pinnacle 21 · Phoenix · the EDC / CTMS / eTMF — never an LLM.
- 3. A human approves any change to a regulated record or timeline**
via the approval queue, with a 21 CFR Part 11 audit trail.

Smartsheet runs the plan; the AI keeps it current

12 automations. Smartsheet's engine computes the critical path; the AI explains, prioritizes, and proposes — the PM approves before any committed-date write.

Critical-path tracking

Nightly + on any change: recompute float, flag tasks now driving the finish, explain what moved.

Data-cut / lock cascade

On the event: recompute downstream dates, draft 'you're now blocked' alerts, post to the approval queue.

Status reports

Weekly/monthly: drafted in the house template from the plan-of-record — numbers from systems of record.

Action & decision capture

From email/meetings (local model): extract owners, due dates, decisions; de-dupe against the log.

Filing, completeness, and SOP knowledge

12 automations. Filing to the record-of-truth is human-approved; RAG answers over your SOPs are cited and advisory.

Auto-classify & file

Propose TMF Reference Model zone/artifact for each document; flag likely duplicates → controller approves.

Completeness & expiry

Detect missing TMF artifacts and expiring CVs / training / licenses; alert owners.

SOP / WI ingestion (RAG)

Approved SOPs embedded into the local store for grounded, cited compliance Q&A.

Inspection readiness

Gap analysis against the TMF model; PII/redaction check before any cloud use.

Recurrent QC triage at every stage

12 automations. The validated tool produces pass/fail; the AI makes it usable — surfacing only what's new.

1

Nightly build — A new SDTM / ADaM dataset lands

2

Pinnacle 21 runs — Validated tool → authoritative Error / Warning / Notice list

3

Local agent triages — Clusters by rule, diffs vs last clean run → only NEW findings; drafts disposition from the cSDRG/ADRG

4

Routed + audited — Each cluster sent to the right programmer; disposition log ready for the reviewer guide

It never changes a verdict. A 4-hour manual triage becomes a clean, pre-explained queue at 8 a.m. — participant-level data stays on the local model.

Risk-based quality management, signalled

12 automations under ICH E6(R3). Unblinded signals run on the local model; reported safety numbers come from validated tools with medical review.

KRI / QTL breach signals

Watch key risk indicators & quality tolerance limits; flag threshold breaches with context.

Central / statistical monitoring

Site outliers, digit preference, enrollment & visit anomalies surfaced for review.

Deviation surveillance

Detect & classify protocol deviations; route for adjudication; track to closure.

DSMB / safety packet assembly

Assemble the review packet; summarize MVRs; track follow-ups — medical review owns the call.

One loop, 75 automations

- 1 Trigger** schedule or study event
- 2 Gather context** connectors (read) + RAG (cited)
- 3 Classify & route** sensitivity → local or cloud, via gateway
- 4 Reason** draft / classify / triage / summarize
- 5 Propose** to the human-approval queue + rationale + sources
- 6 On approval → write** to the system of record, via connector
- 7 Audit** every step → the Part 11 trail

Start small, earn trust

PHASE 0 The stack	Gateway • local model • Claude (BAA) • Presidio guardrail • approval queue • Part 11 trail
PHASE 1 The knowledge base	Ingest SOPs/WIs → ship the SOP/compliance Q&A agent first (pure GREEN, cited, instant value)
PHASE 2 Read connectors + scheduled agents	Smartsheet / eTMF / Pinnacle 21 (read) → nightly QC-triage + weekly status (AMBER, approval queue)
PHASE 3 Event-driven + monitoring	Timeline cascade • eTMF filing proposals • KRI/QTL signals — each graduates as it earns a track record

Each new agent goes live behind the human-approval queue and graduates to lighter review as validation and trust accrue.

WORKED EXAMPLE

A day in the ops layer

02:00

Nightly QC: 3 genuinely new findings surfaced among 240; the rest pre-explained, routed to programmers.

10:30

Data cut declared → critical path recomputed, 'you're blocked' alerts drafted → PM approves → Smartsheet updates in minutes.

13:00

A KRI breaches → the local agent drafts the site follow-up and logs the RBQM register; medical review owns the call.

15:00

3 reports filed to the eTMF on approval; a programmer gets an SOP answer, cited, in seconds.

16:00

The weekly status report drafts itself from the plan-of-record; the PM edits two lines and sends.

75 governed automations. Humans keep ownership.

- 1 Absorbs the cross-system busywork** QC triage, filing, status, tracking, monitoring — across PM, trial management, and monitoring.
- 2 Validated tools + humans own what's regulated** Every reported number and record of truth; PHI stays on the local model.
- 3 Safe by construction** One gateway, one audit trail, a human-approval queue on every regulated change.
- 4 Start with the knowledge base** Ship the SOP Q&A agent first; keep everything behind the queue; expand as trust grows.

The full catalog — every automation, engine, integration, and human checkpoint — is in the operating wiki.